Two-Minute Interview Narration

A concise, technically accurate explanation for interviews.

Project: CreditWise – Loan Approval Prediction

Source: supplied Loan-Approval-main.zip | Notebook: credit_wise.ipynb | Dataset: loan_approval_data.csv

Narration (approximately two minutes)

“I worked on a CreditWise loan approval prediction project. The objective was to estimate whether a loan application would be approved using applicant income, co-applicant income, credit score, debt-to-income ratio, savings, collateral, loan details, employment and other applicant attributes. This is a supervised binary classification problem, with Loan_Approved as the target.”

“The supplied dataset had 1,000 rows and 20 columns. During initial inspection I found 50 missing values in every column, including the target. The notebook filled numeric columns with their mean and categorical columns with their most frequent value. It then removed Applicant_ID, label-encoded Education_Level and the target, one-hot encoded six categorical predictors, and standardized the features.”

“I compared three classifiers: Logistic Regression, K-Nearest Neighbours with five neighbours, and Gaussian Naive Bayes. On the first experiment, Logistic Regression and Naive Bayes both achieved 86.5 percent accuracy. Naive Bayes had the highest precision, around 80.4 percent, while Logistic Regression had an F1 score around 0.777. KNN performed lower, with 76 percent accuracy.”

“I then experimented with squared DTI and credit-score features. In that second run, Logistic Regression achieved 88 percent accuracy, 83.6 percent recall and an F1 score of about 0.81. Gaussian Naive Bayes achieved the highest precision at about 81.1 percent, while KNN remained lower. So the preferred model depends on whether the business places more emphasis on precision or recall.”

“An important limitation I identified is that the notebook imputes missing values before the train-test split and also fills missing Loan_Approved labels with the majority class. That creates leakage and artificial target labels. I would correct this by excluding rows with unknown outcomes, fitting all preprocessing only on training folds, using stratified cross-validation and evaluating fairness, calibration and the costs of false approvals versus false rejections before drawing any operational conclusion.”

Likely interview questions

1. Why Logistic Regression? It is a strong, interpretable baseline that estimates class probabilities and provides a linear decision boundary in transformed feature space.

2. Why scale? Applicant income, savings, credit score and ratios have different units. Scaling is especially important for KNN and often helpful for regularized Logistic Regression.

3. Why KNN underperformed? It is sensitive to noisy or irrelevant dimensions, distance concentration, feature scaling and the choice of K.

4. What is the Naive Bayes assumption? Features are conditionally independent given the class, and GaussianNB models each continuous feature with a class-conditional Gaussian distribution.

5. Why is target imputation wrong? A missing approval outcome is unknown, not a negative decision. Filling it with No creates fabricated labels and biases evaluation.

6. What is data leakage here? Imputation and encoding are fitted on the entire dataset before the split, allowing test-set information to influence preprocessing.

7. Why not use accuracy alone? Error costs are asymmetric; false approvals and false rejections have different financial and social consequences.

8. Which model would you select? Based on the reported single-split results, Logistic Regression has the best second-pass F1/recall, while Naive Bayes has the best precision. I would decide only after leakage-safe CV, threshold/cost analysis and fairness review.

9. What does DTI mean? Debt-to-income ratio, typically recurring debt obligations relative to income; the exact definition and units should be confirmed from the data owner.

10. What would you improve first? Fix target missingness and leakage, then use a pipeline, stratified validation, calibration, cost-sensitive threshold selection and fairness assessment.